

ENVS 193DS Spring 2026 Final

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[Link to GitHub repository](#)

Setup

```
# loading packages
library(tidyverse)
library(janitor)
library(gt)
library(here)

# reading in data
nests <- read_csv(here("data", "nest_data_final.csv"))
```

Problem 1. Research writing

a.

In part 1, they likely used a logistic regression because they were testing whether distance to the water edge predicts the probability of American Avocet microhabitat use.

In part 2, they likely used a chi-square test of independence because they were testing whether there is a relationship between bird species and habitat type, which are both categorical variables.

b.

For part 1, the probability of American Avocet microhabitat use changed with distance from the water edge, suggesting that distance to water may influence where American Avocets choose to use habitat.

For part 2, American Avocets, Forster's Terns, and Black-necked Stilts used habitats differently, suggesting that each species may have different habitat preferences within the San Francisco Bay.

c.

One additional variable that could influence the probability of American Avocet microhabitat use is water depth (cm), which could be measured at each bird observation location using a meter stick or depth gauge. Water depth likely influences habitat use because American Avocets are wading birds that feed in shallow water, and the paper found that they were commonly associated with wet bare ground and shorelines where water is typically shallower.

d.

Problem 2. Data analysis

a.

```

# cleaning and wrangling nest data
nests_clean <- nests %>%

  # selecting columns of interest
  select(case_control, height_cm, ant_species) %>%

  # keeping only the three ant species needed for analysis
  filter(ant_species %in% c("TP", "AB", "RRB")) %>%

  # creating a column with full scientific names
  mutate(
    ant_species_full = case_when(
      ant_species == "TP" ~ "C. sjostedti",
      ant_species == "AB" ~ "C. mimosae",
      ant_species == "RRB" ~ "C. nigriceps"
    )
  ) %>%

  # reordering ant species levels
  mutate(
    ant_species_full = factor(
      ant_species_full,
      levels = c(
        "C. sjostedti",
        "C. mimosae",
        "C. nigriceps"
      )
    )
  )

# displaying the structure of the cleaned data frame
str(nests_clean)

```

```

tibble [234 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control      : num [1:234] 1 0 0 0 0 1 0 0 0 0 ...
 $ height_cm         : num [1:234] 392 310 513 154 324 ...
 $ ant_species       : chr [1:234] "RRB" "RRB" "RRB" "RRB" ...
 $ ant_species_full: Factor w/ 3 levels "C. sjostedti",...: 3 3 3 3 2 3 3 2 3 3 ...

```

```

# displaying 10 random rows from the cleaned data frame
slice_sample(nests_clean, n = 10)

```

```
# A tibble: 10 x 4
  case_control height_cm ant_species ant_species_full
      <dbl>      <dbl> <chr>      <fct>
1         0       232   RRB      C. nigriceps
2         0      136.   RRB      C. nigriceps
3         1      389   RRB      C. nigriceps
4         0      157   RRB      C. nigriceps
5         1      249   RRB      C. nigriceps
6         0      194   RRB      C. nigriceps
7         0      109   RRB      C. nigriceps
8         0      159   RRB      C. nigriceps
9         1      191   RRB      C. nigriceps
10        0       97.5 RRB      C. nigriceps
```

b.

In this data set, a value of 1 indicates that a bird nest was present in the tree, while a value of 0 indicates that no bird nest was present. Biologically, this means the response variable represents whether or not a tree was used as a nesting site by birds.

c.

Tree height: Tree height is a continuous variable measured in centimeters. As tree height increases, I would expect the probability of nest occurrence to increase because taller trees may provide birds with better nesting sites and more protection from predators. I found this information in the Introduction section where the authors discuss how habitat characteristics can influence nesting behavior.

Acacia ant species: Acacia ant species is a categorical variable with three levels: *Crematogaster sjostedti*, *Crematogaster mimosae*, and *Crematogaster nigriceps*. Different ant species may influence nest occurrence because they vary in their behavior and aggressiveness, which could make some trees more attractive or safer for nesting birds than others. I found this information in the Introduction section where the authors describe how species interactions can affect bird nesting behavior.

d.

Model Number	Predictors	Interaction Term?	Model Description
Model 1	Tree height, ant species	No	Tests whether tree height and ant species influence the probability of nest occurrence.
Model 2	Tree height, ant species	Yes	Tests whether the effect of tree height on nest occurrence differs among ant species.